

Real vs. Fake Face Detection — Comprehensive Results Report

Master 2 Thesis — Deepfake Detection Track — Phases 1-7 Complete

1. Objective

Build and validate an end-to-end pipeline for digital face forgery (deepfake) detection, then progressively evaluate and improve it following the roadmap agreed with the supervisor: (1) deeper fine-tuning, (2) cross-dataset generalization evaluation, (3) ensembling, (4) adversarial robustness — extended with two further self-directed phases testing the fixability and extent of the generalization gap found. This report consolidates all seven phases of the experimental work.

2. Dataset & Setup

- Primary dataset (Phases 1, 2, 4, 5, 6): "140k Real and Fake Faces" (Kaggle) — real photographs (FFHQ) vs. StyleGAN-generated faces.
- Cross-domain dataset (Phases 3, 6): "Deepfake and Real Images" (Kaggle) — real face-swap deepfake video frames.
- Third-domain dataset (Phase 7): "Face Dataset Using Stable Diffusion v.1.4" (Kaggle) — Stable Diffusion-generated faces, paired with held-out real photos from the primary dataset.
- Base architectures: ResNet18 and EfficientNet-B0, both pretrained on ImageNet. Training: Colab T4 GPU, Adam optimizer, batch size 64.

3. Phase 1 — Baseline (frozen backbone, linear probe)

- Only the final classification layer trained (1,026 parameters); entire ResNet18 backbone frozen at ImageNet weights.
- 5 epochs, learning rate 1e-3, ~31 minutes total training time.
- Result: 80.8% test accuracy, 0.888 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	8050	1950
True: Real	1896	8104

4. Phase 2 — Deeper Fine-Tuning (unfreeze layer4)

Rationale: Phase 1 plateaued quickly (~80% by epoch 1) because a single linear layer on frozen ImageNet features has limited capacity to specialize to forgery-specific artifacts. Phase 2 unfreezes the last residual block (layer4) of ResNet18, allowing the network's highest-level convolutional features to adapt directly to the task, while earlier layers (generic edges/textures/colors) remain frozen.

- 8,394,754 trainable parameters (vs. 1,026 in Phase 1).
- 5 epochs, learning rate 1e-4 (lowered vs. Phase 1 to protect pretrained weights given the much larger trainable capacity).
- Result: 98.8% test accuracy, 0.999 AUC.

Confusion matrix (n = 20,000):

	Pred: Fake	Pred: Real
True: Fake	9909	91
True: Real	146	9854

5. Phase 1 vs. Phase 2 — Comparison

Metric	Baseline #1 (frozen backbone)	Phase 2 (fine-tuned layer4)
Trainable parameters	1,026	8,394,754
Test Accuracy	80.8%	98.8%
Test AUC	0.888	0.999

Metric	Baseline #1 (frozen backbone)	Phase 2 (fine-tuned layer4)
Precision / Recall / F1 (avg)	0.81 / 0.81 / 0.81	0.99 / 0.99 / 0.99

6. Phase 3 — Cross-Dataset Evaluation (Generalization / Drift)

Both trained models were evaluated, without any retraining, on an independent dataset built from real face-swap deepfake videos, a different manipulation family from the StyleGAN full-face synthesis used for training. Test set size: 10,905 images.

Model	In-dist. Acc / AUC	Cross-dataset Acc / AUC
Baseline #1 (frozen backbone)	0.808 / 0.888	0.487 / 0.489
Phase 2 (fine-tuned layer4)	0.988 / 0.999	0.495 / 0.525

Interpretation

- Both models collapse to near-chance performance on the unseen dataset (AUC 0.489 and 0.525, vs. 0.888 and 0.999 in-distribution).
- Both models are heavily biased toward predicting "Real" on this unseen data rather than failing randomly.
- This constitutes the thesis's first empirical generalization/drift result, directly evidencing the cross-generator generalization gap identified as the central open challenge in Section 8 of the state-of-the-art review.

7. Phase 4 — Ensemble (ResNet18 + EfficientNet-B0)

A second, independently trained architecture (EfficientNet-B0, last feature block + classifier head fine-tuned) was combined with the Phase 2 ResNet18 via simple probability averaging, and evaluated on both the in-distribution and cross-dataset test sets.

Model	In-dist Acc	In-dist AUC	Cross Acc	Cross AUC
ResNet18 (Phase 2)	98.81%	0.999	49.45%	0.525
EfficientNet-B0	93.82%	0.987	48.88%	0.499
Ensemble (avg)	98.99%	0.999	49.35%	0.498

Interpretation

- On the cross-dataset test, the ensemble's AUC (0.498) is slightly lower than ResNet18 alone (0.525) — EfficientNet-B0's cross-dataset AUC is essentially exact chance (0.499), so averaging pulls the combined result toward chance rather than improving it.
- This demonstrates that architectural diversity alone does not resolve the generalization gap: both models share the same single-source blind spot, so ensembling combines two instances of the same failure mode.

8. Phase 5 — Adversarial Robustness

The Phase 2 fine-tuned ResNet18 was evaluated against two standard white-box adversarial attacks — FGSM (single-step) and PGD (10-step iterative) — on a 1,000-image evaluation subset, across a range of perturbation strengths (epsilon).

Epsilon	FGSM Accuracy	PGD Accuracy
0.00 (clean, no attack)	99.0%	99.0%
0.01	0.0%	0.0%
0.03	0.1%	0.0%
0.05	13.2%	0.0%
0.10	44.9%	0.0%

Interpretation

- At epsilon = 0.01 — a near-imperceptible perturbation — accuracy collapses to 0% under both attacks. PGD remains at 0% across all tested epsilon values.

- FGSM's non-monotonic behavior (0% -> 44.9% as epsilon increases further) is a documented property of its single-step linear approximation breaking down at larger epsilon, not observed with the more reliable PGD attack.
- This characterizes the undefended model's baseline vulnerability, expected to be high for any standard CNN not specifically trained for robustness.

9. Phase 6 — Mixed-Source Training

Motivated by the Phase 3 and Phase 4 findings, this phase tests whether the generalization gap is fixable through training data diversity. A fresh ResNet18 was fine-tuned (identical recipe to Phase 2) on a combined training set drawn from both the StyleGAN and face-swap sources (30,000 real + 30,000 fake training images, balanced across both sources), then evaluated on held-out test sets from each source individually.

Model	Test Set	Accuracy	AUC
Phase 2 (single-source)	StyleGAN	98.68%	0.999
Phase 2 (single-source)	Face-swap	49.78%	0.525
Phase 6 (mixed-source)	StyleGAN	95.49%	0.995
Phase 6 (mixed-source)	Face-swap	82.42%	0.928

Interpretation

- On the face-swap test set — where Phase 2 was essentially blind (49.8% accuracy, 0.525 AUC) — the mixed-source model achieves 82.4% accuracy and 0.928 AUC, closing roughly 85% of the total possible gap between chance-level and near-ceiling performance.
- On the StyleGAN test set, the mixed-source model shows only a small reduction relative to Phase 2 (95.5% vs. 98.7% accuracy) — strong performance on the original domain is retained while substantial competence is gained on the new one.
- This directly confirms that the generalization failure was primarily a consequence of training data diversity, not a fundamental architectural limitation, and constitutes the thesis's strongest positive empirical result to this point.

10. Phase 7 — Third-Domain Generalization Test

Phase 6 showed strong generalization gains on the two forgery types it was trained on. This phase tests a stricter, more skeptical question: does that benefit extend to a completely unrelated, unseen third forgery type, or is it specific to the two types trained on? Both the Phase 2 and Phase 6 models were evaluated, without retraining, on Stable Diffusion-generated faces (a fundamentally different generation paradigm from both GANs and face-swapping), paired with held-out real photographs never used in training. Note: due to the third-party dataset's size, this evaluation set is class-imbalanced (2,536 fake / 5,000 real), so AUC (threshold-independent) is the primary reliable metric; accuracy is reported for completeness only.

Model	Accuracy*	AUC
Phase 2 (single-source)	65.5%	0.277
Phase 6 (mixed-source)	66.4%	0.552

Interpretation

- Phase 2's AUC (0.277) is notably below 0.5, indicating a systematic inverse relationship: the single-source model rates diffusion-generated fakes as more likely to be real, on average, than it rates actual real photographs. The confusion matrix confirms this — only 4 of 2,536 diffusion fakes were correctly identified (a 99.8% miss rate). This suggests the StyleGAN-only model learned a feature correlated with "real" (e.g. image smoothness or cleanliness) that diffusion-generated images satisfy even more strongly than genuine photographs.
- Phase 6's AUC (0.552) is a real, measurable improvement over Phase 2 — correcting the inverse relationship and reaching just above chance level — but remains far short of reliable detection (its fake catch rate improved from 0.2% to 14.6%, still low in absolute terms).
- Overall, this is a more guarded finding than Phase 6's headline result: mixed-source training (on StyleGAN + face-swap) provides a partial, measurable benefit even on a completely unrelated third generation method (diffusion), but does not achieve reliable open-set generalization to an arbitrary unseen generator. This is a more precise and defensible

claim than "mixed-source training solves generalization," and is consistent with the broader literature's view of cross-generator generalization as a substantially unresolved problem (SOTA Section 8).

11. Overall Conclusion (Phases 1-7)

Across seven phases, this work establishes a complete and nuanced empirical narrative. A fine-tuned ResNet18 achieves excellent in-distribution performance (98.8% accuracy, 0.999 AUC) when trained and tested on a single generator family (Phases 1-2), but this does not reflect a generally robust deepfake detector: the model collapses to near-chance accuracy on an unseen face-swap dataset (Phase 3), and ensembling with an independently trained EfficientNet-B0 does not recover this loss (Phase 4), since both models share the same single-source blind spot. The model additionally exhibits near-total vulnerability to imperceptible adversarial perturbations (Phase 5). Diversifying the training data to include two forgery types (Phase 6) recovers the large majority of lost generalization performance on those two types (Face-swap AUC: 0.525 $\rightarrow$ 0.928) while retaining strong in-distribution performance — the thesis's strongest positive result. However, this benefit only partially transfers to a third, unrelated forgery paradigm (diffusion-generated faces, Phase 7): AUC improves from a below-chance 0.277 to a modest 0.552, correcting a systematic bias but falling well short of reliable detection. Together, these results provide direct empirical evidence for the central open challenge identified in the thesis's state-of-the-art review (Section 8), and additionally show that training data diversity, while substantially more effective than architectural ensembling at closing the generalization gap for known forgery types, does not by itself solve open-set generalization to arbitrary unseen generators. Adversarial robustness (Phase 5) also remains an entirely open weakness, unaddressed by mixed-source training.

12. Next Steps

- Adversarial training (or other defense) to address the Phase 5 robustness gap, which mixed-source training did not target.
- Investigate the Phase 7 below-chance AUC finding further (e.g. feature-level analysis of what the single-source model associates with "real") as a specific, well-defined sub-question.
- Consider frequency-domain features (SOTA Section 4.2) or a foundation-model-based approach (e.g. CLIP adapter, SOTA Section 4.4), which may generalize more broadly across generation paradigms than a plain fine-tuned CNN, even with mixed-source training.
- If pursuing broader open-set generalization further, consider domain-generalization techniques explicitly designed for this setting (e.g. adversarial domain discriminators, meta-learning-based approaches) rather than simple data mixing alone.

Code & results (Colab notebooks, outputs included): [\[GitHub repository link\]](#)